

Unit 2 Mini-Cases

Principles of Capital Markets

Winter Term 2025/26

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **October 22, 2025 at 6am**. To do so, go to the assignment section on Canvas and choose “Unit 2 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Questions 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini-Case 2.1 (pdf)

You have the choice of investing into two companies, A and B, with the following return distributions:

Company	Probability	Return
A	0.25	0.20
	0.65	0.05
	0.10	-0.3125
B	0.70	0.1375
	0.30	-0.15

Compute the variance and standard deviation for the returns of both companies.

Mini-Case 2.2 (pdf)

Investors A and B both hold stocks worth either 15,000 EUR, or 7,000 EUR, with equal probabilities, in the next period. Investor A and B have the same type of utility function $U(W) = \frac{W^{1-\gamma}-1}{1-\gamma}$. Investor A's γ is 0.75, Investor B's γ is 0.5.

- (a) Calculate the expected wealth and the expected utility of wealth for each investor.
- (b) What is the lowest amount the investors are willing to sell their positions for right now instead of holding them until the next period?
- (c) If the investors could sell their positions now for 10,500 EUR, would they decide to do so?

Mini-Case 2.3 (pdf)

Consider the below returns over four periods for two securities, A and B.

Period	Return Stock A	Return Stock B
1	0.07	0.10
2	0.10	0.04
3	0.005	-0.05
4	0.06	0.01

- (a) Compute the covariance between security A and B manually.
- (b) Determine the magnitude of comovement between security A and B by calculating their correlation. Discuss your results.

Mini-Case 2.4 (Excel or R)

You wish to diversify your investment portfolio and have therefore decided to invest in four US stocks. Download stock price data for Oracle (ORCL), Chevron (CVX), Target (TGT) and Bank of America (BAC) provided in the assignment section of Unit 2 ($\rightarrow$ "StockDataUnit2.zip"). Use monthly closing prices (column "PRC") for the entire observation period for your analyses. Assume that historical data can be used to estimate expected values of future data.

- (a) Calculate the monthly returns for each stock.
- (b) Calculate the expected return and return variance for each stock.
- (c) Calculate the annualized expected return, annualized variance, annualized standard deviation for each stock. Interpret your results.

Hint: You can make use of the following approximations: $E[\tilde{r}_{annual}] \approx 12 \times E[\tilde{r}_{monthly}]$ (neglecting compounding) and $var(\tilde{r}_{annual}) \approx 12 \times var(\tilde{r}_{monthly})$ (assuming independence).

Mini-Case 2.5 (Excel or R)

Use the information given in *Mini-Case 2.4*.

- (a) Calculate the covariance matrix for the four stocks.
- (b) Calculate the correlation matrix for the four stocks.
- (c) Interpret your results.

Mini-Case 2.6 (pdf)

You have the choice of investing your wealth into a Portfolio P consisting of any strictly positive combination of (i) a risk-free bond with return r_f and (ii) a risky stock with random return $\tilde{r}_s$. Let's denote your proportional investment into the risk-free bond by $0 \leq x_1 \leq 1$ and your proportional investment into the risky stock by x_2 . Furthermore, let's assume that a rational investor's investment decisions can be explained by the following (expected) utility function:

$$E[U(\tilde{R}_P)] = E[\tilde{R}_P] - \frac{\gamma}{2} E[(\tilde{R}_P - E[\tilde{R}_P])^2] = E[\tilde{R}_P] - \frac{\gamma}{2} \text{var}(\tilde{R}_P),$$

where $\tilde{R}_P$ denotes the random return of Portfolio P .

- (a) Derive your optimal investment into the risky stock x_2^* by maximizing the above utility function.

Hint: First, write out $\tilde{R}_P$ in the above equation. Then, take the usual first order condition of a simple optimization problem. Remember that $x_1 = 1 - x_2$ and $\text{var}((1 - x_2)r_f + x_2\tilde{r}_s) = x_2^2 \text{var}(\tilde{r}_s)$.

- (b) How does your risk aversion γ influence x_2^* ?